Computing and Artificial Intelligence Cluster
Capstone Project Workbook



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Supervisor	MS. Dina Sarris
Capstone Project Title	Shopwise AI Cart
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Section1- Grade 11



G11 Term 3 – Project Overview

The CAI Capstone Project gives you the opportunity to apply the skills and knowledge you have learned at ATS to solve a real-world problem using AI technologies. You will design, develop, test, and improve a digital or AI-based solution while demonstrating creativity, problem-solving, and technical skills.

Throughout the project, you will plan your work, document your progress, and collaborate with your team, similar to how professionals work in technology fields. The project encourages innovation and the use of modern AI tools and technologies to create a high-quality final product.

Outstanding projects may be showcased at the school's TVET event. This capstone project is your chance to demonstrate your readiness for future studies and careers in computing and AI while creating a project you can participate in multiple showcase competitions and innovation awards.

G11 Term 3- Project Planning and Proposal Phase

Term 3 Weekly Plan: Project Planning and Proposal
This term, you will start planning your Capstone Project, which you will fully implement and complete in Grade 12, Term 1. You will explore project ideas, investigate different technical approaches, and develop a clear concept that is both practical and innovative. You will work like a professional AI /Software engineer, considering the end-user and market need in your plan while exploring the recent AI tools and technologies for optimal utilization in your project. This ensures your ideas are realistic, well-organized, and ready to be brought to life. By the end of week two, you will create a complete project proposal. This proposal will explain what you plan to build, why it is important, and how you will carry it out. It will be submitted to the ATS Directorate CDU Specialist for review and approval. Once approved, your proposal will guide all your planning and practical work in Grade 12, Term 1, helping you stay focused and on track to produce a high-quality project.

Week 1–2: Explore and Select Project Concept

Form a project group of no more than four students, ensuring members demonstrate strong communication skills, teamwork, responsibility, time management, problem-solving ability, and commitment, as these soft skills are essential for successful project completion.

- Brainstorm multiple project ideas with guidance from your supervisor teacher.
- Conduct preliminary research to understand real-world applications and existing solutions.
- Discuss ideas with peers and your supervisor to get feedback and advice.
- Select one project idea to develop further.
- Define the project's goal, purpose, expected outcomes, and the problem it aims to solve, with supervisor input.

Week 3–5: Prepare and Submit Project Proposal

- Conduct research on software, tools, technologies, GenAI & Agentic AI principles relevant to your project.
- Develop a detailed project proposal under the guidance of your supervisor, including:
    - Project title and description
    - Problem statement
    - Background research
    - Project scope
    - Platforms, AI tools, software and any resources required
    - Expected outcomes
- Submit the completed proposal to the Computing CDU Specialist by the end of Week 4–5.
- Ensure your supervisor reviews your proposal before submission for accuracy and completeness.



Computing and AI Capstone Project Proposal – Fill out form

Project proposal submissions guidelines

- All team members must actively participate in planning and development tasks under the guidance of the supervisor.
- The supervisor must review and approve the project work before any official submission or progression.

Only one project submission is required per group, representing the collective work of the team.

Week 6–7: Feedback and Revision

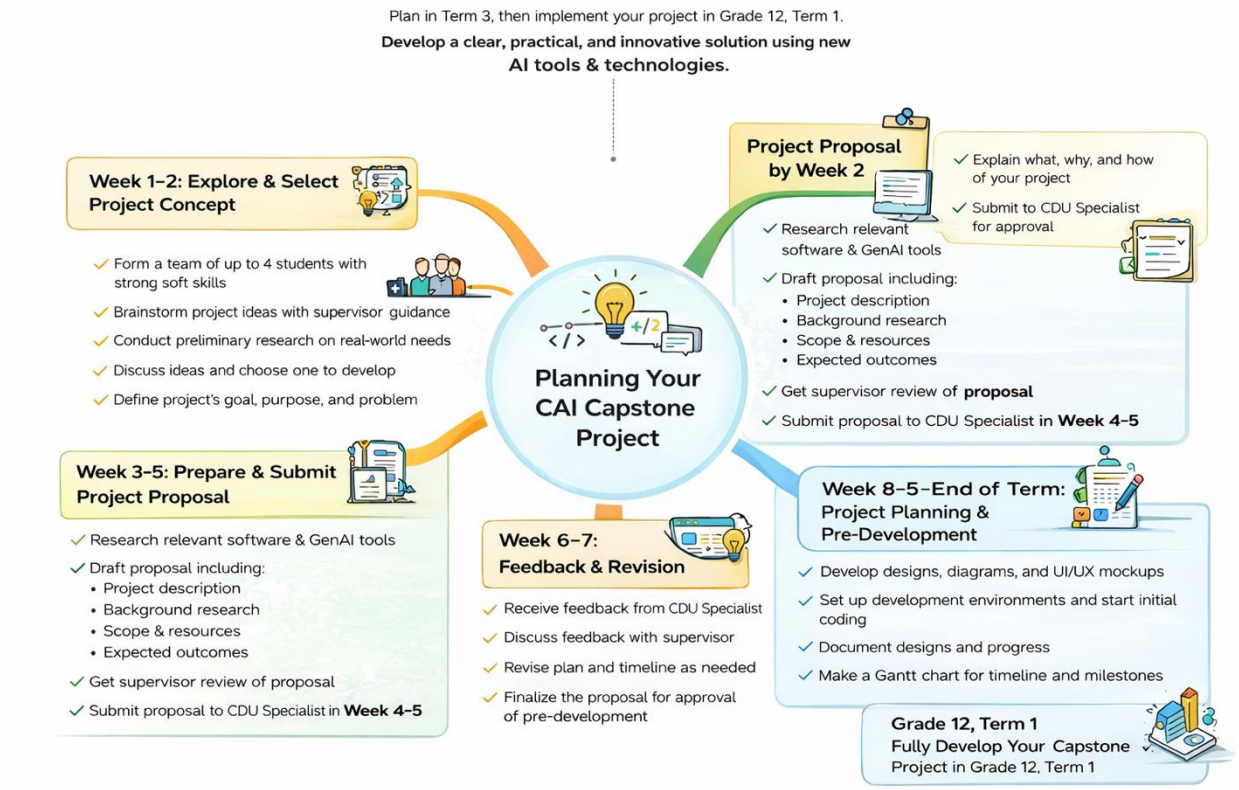
- Receive feedback and suggestions from the CDU Specialist.
- Discuss the feedback with your supervisor to understand required changes.
- Revise your project plan, work flow chart, or timeline accordingly.
- Finalize the approved proposal under supervisor guidance to ensure it is realistic and well-structured for pre-production work.

Week 8–End of Term: Project Planning and Pre-Development

Begin detailed pre-development planning with supervisor support:

- Develop system designs, wireframes, flowcharts, or hardware architecture diagrams.
- Create UI/UX stryboard or interface designs where applicable.
- Design algorithms, data flow diagrams, or AI model structures.
- Identify required software, datasets, tools, platforms, and resources.
- Set up development environments and start preliminary coding or prototyping under supervisor guidance.
- Document all planning, designs, decisions, and progress clearly.

- Develop a Gantt chart to visualize the full project timeline, including:
  - Planning, development, testing, and deployment milestones
  - Key deadlines and supervisor check-ins to monitor progress
- All work in Term 3 focuses on planning and preparation. Full development, testing, documentation, and final presentation will continue in Grade 12, Term 1.



G11 Learner Evidence

Learner evidence Week 1-2	**• Our team members are Dana Alawbthani, Dan Mohammed, Alya Aref, and Rawda Waleed.** **• Dana researched AI technologies and smart shopping applications to support the project idea.** **• Dan researched image recognition and how AI can identify products using a smartphone camera.** **• Alya explored budgeting features, allergy detection systems, and user requirements.** **• Rawda researched store navigation, personalization features, and different diet modes such as halal, vegan, and keto.** **• Together, we brainstormed project ideas and selected Shop Wise AI Cart as our final project.**

	• We identified shopping challenges such as overspending, allergy risks, and food waste, then planned features to solve these problems using AI.
Learner evidence Week 3-5	**• Conducted research on AI technologies, software, and tools required for the project.** **• Developed the project proposal, including the project description, problem statement, scope, and expected outcomes.** **• Designed the application prototype and user interface using Figma.** **• Identified and finalized the main features of Shop Wise AI Cart, such as image recognition, budget tracking, allergy alerts, store navigation, and personalized recommendations.** **• Discussed feedback with the supervisor and made improvements to the project design and proposal.** **• Prepared and submitted the project documentation.**
Learner evidence Week 8-End of Term	**Developed the system design, wireframes, and application structure.** **• Completed the Figma prototype and continued improving the user interface design.** **• Added and planned features such as budget tracking, allergy alerts, diet preferences, store selection, store navigation, and expiry reminders.** **• Set up the development environment and started developing the application in Visual Studio Code using React and TypeScript.** **• Currently working on coding, testing, and improving the application features.** **• Documenting project progress, designs, and development decisions.** **• Preparing the project for the next stage of development and testing.**

Section2- Grade 12



G12 Term 1 overview

Welcome to Term 1 of your CAI Capstone Project!

This term focuses on turning your approved plan into a fully working computing or AI solution. You will develop, test, improve, and finalize your project while applying your programming, problem-solving, and AI skills.

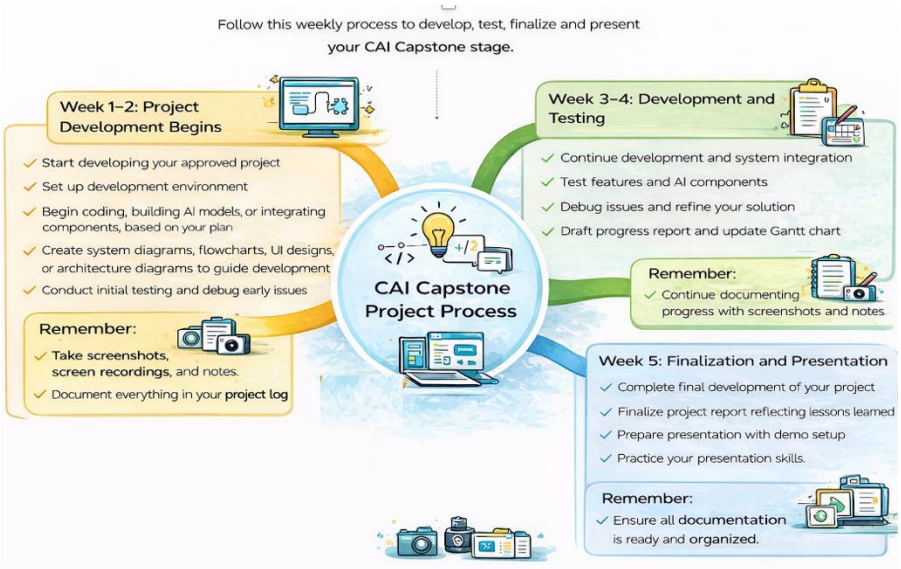
With support from your supervisor, you will manage your time, document your progress, solve technical challenges, and work effectively as a team.

At the end of the term, you will present your project at the Capstone Showcase, demonstrating your solution and reflecting on what you learned. This is your opportunity to showcase your skills, creativity, and innovation.

G12 Term 1- Project Implementation Phase

Term 1 Weekly Plan: Project Production, Testing and Presentation	
	Week 1–2: Project Development Begins • Start developing your approved project under the guidance of your supervisor. • Set up your development environment and project structure. • Begin coding, building AI models, or integrating system components based on your plan. • Create system diagrams, flowcharts, UI designs, or architecture diagrams to guide development. • Conduct initial testing and debug early issues. Remember: Take screenshots, screen recordings, or photos of your progress and keep clear notes in your project log. These will serve as evidence of your work. **Week 3–4: Project Development and Testing** • Continue development, coding, and system integration under supervisor guidance. • Test individual features and AI components to ensure functionality, accuracy, and efficiency. • Debug issues and refine your solution based on testing results.

	• Begin drafting your project report, including methodology, system design, tools used, and problem-solving steps. • Update your Gantt chart to track tasks, milestones, and progress. **Remember:** Continue documenting your work with screenshots, recordings, and notes at every stage to show your development process. **Week 5 – Capstone Project Showcase: Finalization and Presentation** • Complete the final development of your project and ensure all features work as planned. • Finalize your project report, including objectives, methodology, system design, results, challenges, solutions, and lessons learned. • Prepare your presentation materials: slides, live demo setup, visuals, and key talking points. • Practice presenting and demonstrating your solution under supervisor guidance. • Present your project at the Capstone Showcase, highlighting: ▪ The problem your project solves and its purpose. ▪ How your system works and any AI or technical innovations. ▪ Testing results and evaluation outcomes. ▪ What you learned and possible future improvements. **Remember:** Ensure all screenshots, recordings, photos, and documentation are organized and ready, as they form part of your project evidence.

G12 Learner evidence

Appendix

Appendix A- Recommended Technology Resources

The following tools, platforms, and resources are recommended for use in CAI Capstone Projects.

Students should select technologies that match their project goals, technical skills, and supervisor guidance.

The use of modern AI tools is encouraged, as they are used ethically and responsibly.

1- Core Programming & Development Tools

Recommended for all CAI projects.

Category	Recommended Tools
Coding IDEs	Visual Studio Code, PyCharm, Jupyter Notebook, Google Colab, Replit
Programming Languages	Python, JavaScript, SQL
Version Control	GitHub

2- AI & Machine Learning Tools

Used for AI-based solutions and intelligent systems.

Category	Recommended Tools
AI Frameworks	TensorFlow, PyTorch, Scikit-learn, Keras
Computer Vision	OpenCV, MediaPipe, YOLO
NLP & LLMs	OpenAI API, Hugging Face Transformers, IBM Watson
GenAI Tools	ChatGPT, Microsoft Copilot, Google Gemini

3- Web, App & Interface Development

For projects requiring user interaction.

Category	Recommended Tools
UI/UX Design	Figma, Adobe XD, Canva
Web Frameworks	Streamlit, Gradio, Flask, FastAPI, React

Backend & Database	Firebase, SQL Databases

4- Data Science & Analytics

For data-driven and AI training projects.

Category	Recommended Tools
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Plotly, Power BI, Tableau Public
Data Sources	Kaggle, UCI Repository, Hugging Face Datasets

5- Automation & Agentic AI (Advanced Projects)

Recommended for advanced teams.

Category	Recommended Tools
Automation	n8n, Zapier
AI Agents	LangChain, CrewAI, AutoGen

6- Cloud & Deployment Platforms

For hosting and demonstrating final projects.

Category	Recommended Tools
Cloud Platforms	AWS Educate, Google Cloud, Microsoft Azure
Deployment	Render, Railway, GitHub Pages

Appendix B- Suggested Hardware Component

This matrix helps students select suitable hardware components for building AI, IoT, automation, robotics, or smart system prototypes. Hardware selection should always be discussed with the supervisor before implementation.

1- Core Computing & Development Boards

Hardware Component	Primary Use	Project Examples	Skill Level	Notes
Arduino Uno / Nano	Sensor control & embedded systems	Smart sensors, automation prototypes	Beginner	Best for hardware control (not heavy AI processing)
Raspberry Pi	Edge computing, AI processing, IoT systems	Smart home, AI vision, automation systems	Intermediate	Supports Python, AI libraries, camera modules
ESP32 / ESP8266	IoT & wireless communication	IoT monitoring systems, smart devices	Intermediate	Built-in WiFi & Bluetooth
Jetson Nano (NVIDIA)	AI edge computing	Computer vision, real-time AI projects	Advanced	Suitable for deep learning applications

2- AI Vision & Camera Modules

Hardware Component	Primary Use	Project Examples	Skill Level	Notes
USB Webcam	Basic computer vision	AI detection prototypes	Beginner	Easy integration with Python/OpenCV
Raspberry Pi Camera Module	Computer vision & image capture	Object detection, smart surveillance	Intermediate	Works directly with Raspberry Pi
AI Camera / Smart Vision Camera	AI-based visual recognition	Smart monitoring, AI inspection	Advanced	Supports edge AI processing

3- Sensors (IoT & Smart Systems)

Hardware Component	Primary Use	Project Examples	Notes
Temperature & Humidity Sensor	Environmental monitoring	Smart agriculture, smart rooms	Simple data collection
Ultrasonic Sensor	Distance detection	Robotics, smart parking	Common for automation projects
Motion (PIR) Sensor	Movement detection	Smart security systems	Easy integration
Light Sensor (LDR)	Light level detection	Smart lighting systems	Low-cost and simple

Gas / Air Quality Sensor	Environmental AI projects	Smart safety systems	Useful for real-world impact projects

4- Robotics & Automation Components

Hardware Component	Primary Use	Project Examples	Skill Level	Notes
Servo Motors	Controlled movement	Robotic arms, smart mechanisms	Beginner	Easy with Arduino
DC Motors + Motor Driver	Movement systems	Smart vehicles, automation	Intermediate	Requires motor control logic
Robotic Kit	Robotics prototyping	Autonomous robot projects	Intermediate	Combine with sensors + AI
Humanoid Robot Kits	AI interaction projects	Educational robotics	Advanced	Optional for advanced teams

5- Drone & Autonomous Systems

Hardware Component	Primary Use	Project Examples	Skill Level	Notes
Programmable Drone	Autonomous systems	AI navigation, smart inspection	Advanced	Requires safety supervision
Drone Camera Module	Aerial data collection	Vision-based AI projects	Advanced	Combine with AI models

Appendix C- Suggested AI tools & Technology Stack

1- AI Tools Matrix (based on the project type)

Project Type	Suggested Tools
AI Chatbot / Assistant	Python, OpenAI API, Streamlit, LangChain
Data Analysis Project	Python, Pandas, Power BI, Jupyter
Computer Vision Project	Python, OpenCV, YOLO, TensorFlow
Web-Based AI App	React / Streamlit, Flask, AI APIs
Mobile AI App	Swift, CoreML, AI APIs
Automation / Smart Workflow	n8n, AI APIs, Python

2- Technology Stack (based on the project type)

Project Category	Suggested Stack
AI Assistant	Python + OpenAI API + Streamlit
Smart Data Dashboard	Python + Power BI + Pandas
AI Vision System	OpenCV + TensorFlow + Python
Educational AI Tool	Python + Gradio + Hugging Face
Smart Automation System	n8n + AI APIs
AI Web Platform	React + FastAPI + Cloud Deployment